State and Permission Matrix

Two questions on one page: **who may do what** (role and department), and **what may be done in each status** (the locks). Both are checked by the backend on every call; the frontend only hides what a user may not use.

1. Roles

Role	Department	Meaning
Admin	any (only a label)	does everything
Worker	Sales	Sale Worker: quotations (a production order is read only)
Worker	Factory	Factory Worker: the processes, production, stock, purchase

A Worker must have the department `Sales` or `Factory`; the department decides the access. The role check comes before the check that the record exists: a user who may not touch a record gets `403`, never `404`.

2. Who may do what

Area	Action	Admin	Sale Worker	Factory Worker
Dashboard	order tiles	yes	yes	yes
Dashboard	invoice tiles and chart	yes	no	no
QT	see, create, change, print	yes	yes	no
QT	upload customer PO, Confirm, delete a Draft	yes	yes	no
Production order	read	yes	yes	yes
Production order	add or change the processes before the start	yes	no	yes
Production order	Start production (deducts stock)	yes	no	yes
Production order	Record progress	yes	no	yes
Production order	change or reopen a Completed process	yes	no	no
Production order	evidence files: read	yes	yes	yes
Production order	evidence files: upload, delete	yes	no	yes
Stock	read materials, lots, movements	yes	yes	yes
Stock	Add / Deduct / Adjust a lot	yes	no	yes
Materials	create, change, delete	yes	no	no
Purchase	PO: create, change, print, move status, pay, files	yes	no	yes
Suppliers	read	yes	no	yes
Suppliers	create, change, delete	yes	no	no
Customers	read	yes	yes	no
Customers	create, change, delete	yes	no	no

Area	Action	Admin	Sale Worker	Factory Worker
Products	read	yes	yes	yes
Products	create, change, delete	yes	no	no
Accounting	invoices, payments, receipts	yes	no	no
Users	read, create, change, delete	yes	no	no

Menus follow the same table: an entry shows only for the roles and departments that may use it.

3. Statuses and what each allows

QT

Status	Customer and lines	Other fields	Delete	Confirm	Next
Draft	change	change	yes	yes (once)	Confirmed
Confirmed	locked (`409 QT_LOCKED`)	change	no (`409 IN_USE`)	already done: shows the existing order	–

Confirm needs a customer and at least one line. A missing customer PO file is only a warning. Confirm makes one Draft production order and copies the PO file to it.

Production order (status is computed from its processes)

Status	Edit lines and processes	Start	Record progress	Next
Draft, not started	yes (Admin, Factory Worker)	needs a line, a process on every line and enough stock (`409 ORDER_NOT_READY`, `409 STOCK_SHORT`)	no (`409 ORDER_NOT_STARTED`)	In progress
In progress	no (locked by the start)	already started	yes	Completed
Completed	no	–	no, except an Admin reopening a process (note required)	invoiced
Invoiced	no	–	no (`409 ORDER_INVOICED`)	–

Process

Status	Quantity	Who may change
Not started	0	Admin, Factory Worker
In progress	partial	Admin, Factory Worker
Completed	the full quantity	Admin only; reopening needs a note (`403 COMPLETED_LOCKED` for a Worker)

Purchase order

Status	Change lines	Delete	Files	Payment	Next (by hand, one step only)
Draft	yes	yes	yes	no	Sent
Sent	no (`409 PO_LOCKED`)	no	yes	no	Received
Received	no	no	yes	once (`409 ALREADY_PAID`)	–

Received adds the stock once, into the lot typed for each line; a second Received changes nothing (`409 INVALID_STATUS_CHANGE`). Payment before Received answers `409 PO_NOT_RECEIVED`. The amount must equal the total.

Invoice

Status	Payment	Result
Unpaid	one full payment (amount = total)	Paid
Paid	no (`409 ALREADY_PAID`)	receipt issued, once

An invoice is made only for a Completed order (`409 ORDER_NOT_COMPLETED`), one per order (`409 INVOICE_EXISTS`), as a snapshot. It locks the progress of the order.

Supplier (the only record with an Active switch)

Active	New or changed PO	Existing POs
yes	can be chosen	unchanged
no	cannot be chosen	stay as they are

4. One job through the statuses

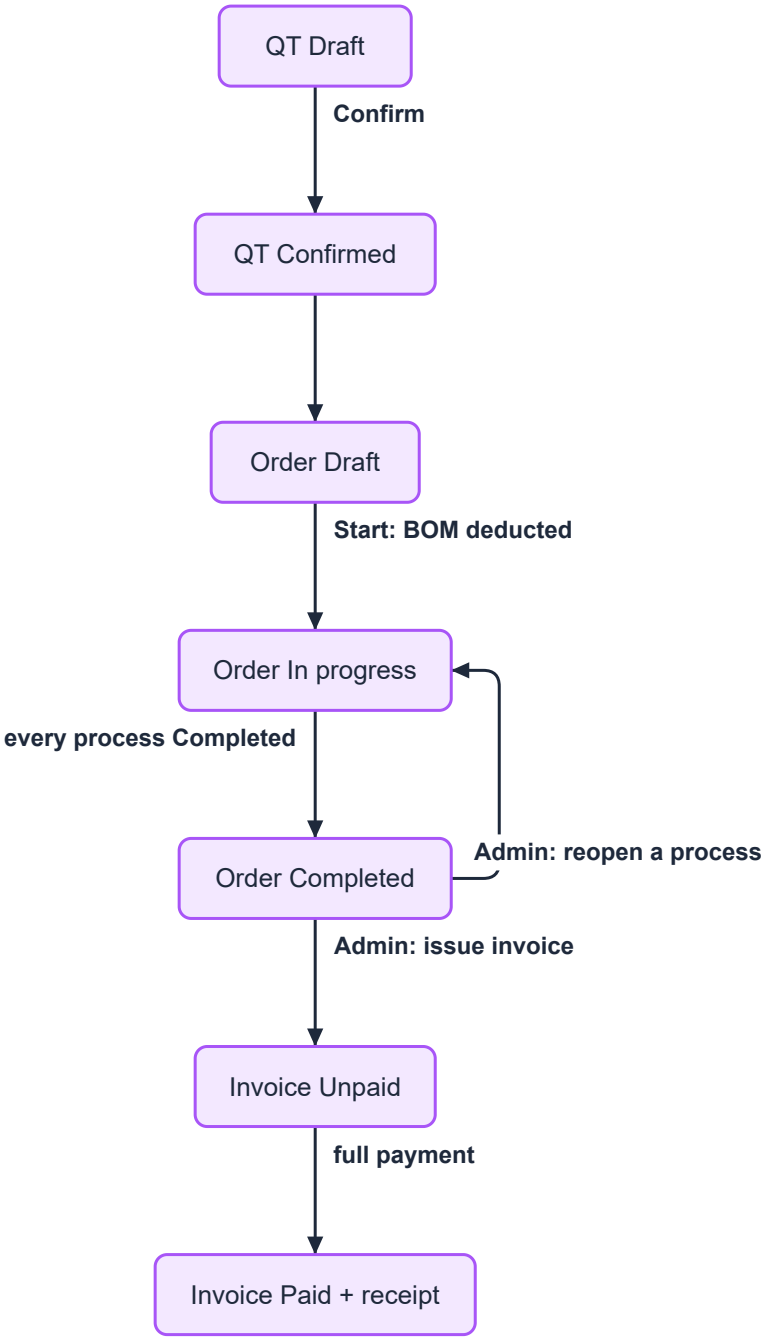


Figure 1. 4. One job (QT, order, invoice) and one purchase order through the statuses

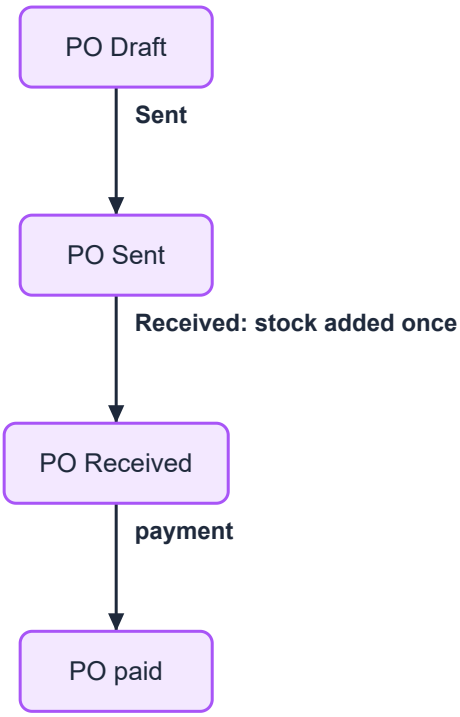


Figure 2. 4. One job (QT, order, invoice) and one purchase order through the statuses

5. Locks at a glance

Lock	Starts when	What it blocks	Error
QT locked	Confirm	change of customer and lines, delete	`QT_LOCKED`, `IN_USE`
Order locked	Start production	change of lines and processes	`ORDER_LOCKED`
Completed process	process Completed	change by a Worker	`COMPLETED_LOCKED`
Order invoiced	invoice issued	progress updates	`ORDER_INVOICED`
PO locked	status leaves Draft	change and delete of the PO	`PO_LOCKED`
PO file copy	Confirm QT	delete of the PO file copy on the order	`FILE_LOCKED`
In use	another record uses it	delete of customers, products, materials, suppliers	`IN_USE`